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MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Mobile Applications for IOS

Multimedia data compression. Compression formats. Video and audio streaming.

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PROGRAMME CONTENT		
Lecture		Number of hours
Lec 1	Introduction. Review of selected mobile applications. Introduction to Apple devices and Mac OS.	2
Lec 2-3	UI and UX. Interactions. Touch screens. Material design. Human Interface guidelines. Voice communication. Siri.	4
Lec 4	Tools for multimedia processing. Gathering multimedia data. Copyright.	2
Lec 5	Implementation rules in Xcode. The structure of application written in Swift.	2
Lec 6-7	Presentation of basic elements of Swift by examples. Core libraries.	4
Lec 8	Apple Human Interface Guideline by examples.	2
Lec 9	Libraries and frameworks supporting creation of multimedia applications. Short characteristics of Kotlin language.	2
Lec 10	Applications of augmented reality. Code analysis of application with augmented reality mechanisms.	2
Lec 11	Games review. Connections between games and the progress of algorithms and programming languages.	2
Lec 12	Multimedia data compression. Compression formats. Video and audio streaming.	2
Lec 13	Multimedia in mobile systems. Cameras. Recommended frameworks.	2
Lec 14	Interaction mechanisms – review. New technologies and multimedia devices.	2
Lec 15	Summary. Perspectives of multimedia techniques.	2
	Total hours	30



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Announcements



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Upcoming Schedule

16.12.2025 - Lecture number 10

13.01.2026 - Lecture number 11 and last attendance check.

20.01.2026 - Lecture number 12 and deadline for your essay (students who currently have **more than one absence**)

27.01.2026 - Written exam

03.02.2026 - Written exam (retake)



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All students who would like to transfer their laboratory grade to the lecture grade, but currently have **more than one absence**, may make up for the missing classes by preparing an **essay in the form of a presentation or a written document** on one of the following topics:

- Multimedia in mobile systems
- Interaction mechanisms, new technologies, and multimedia devices
- Perspectives and future trends in multimedia technologies
- Mobile e-commerce: trends, technologies, and user experience
- Other related topics (subject to approval)

Deadline: 20.01.2026



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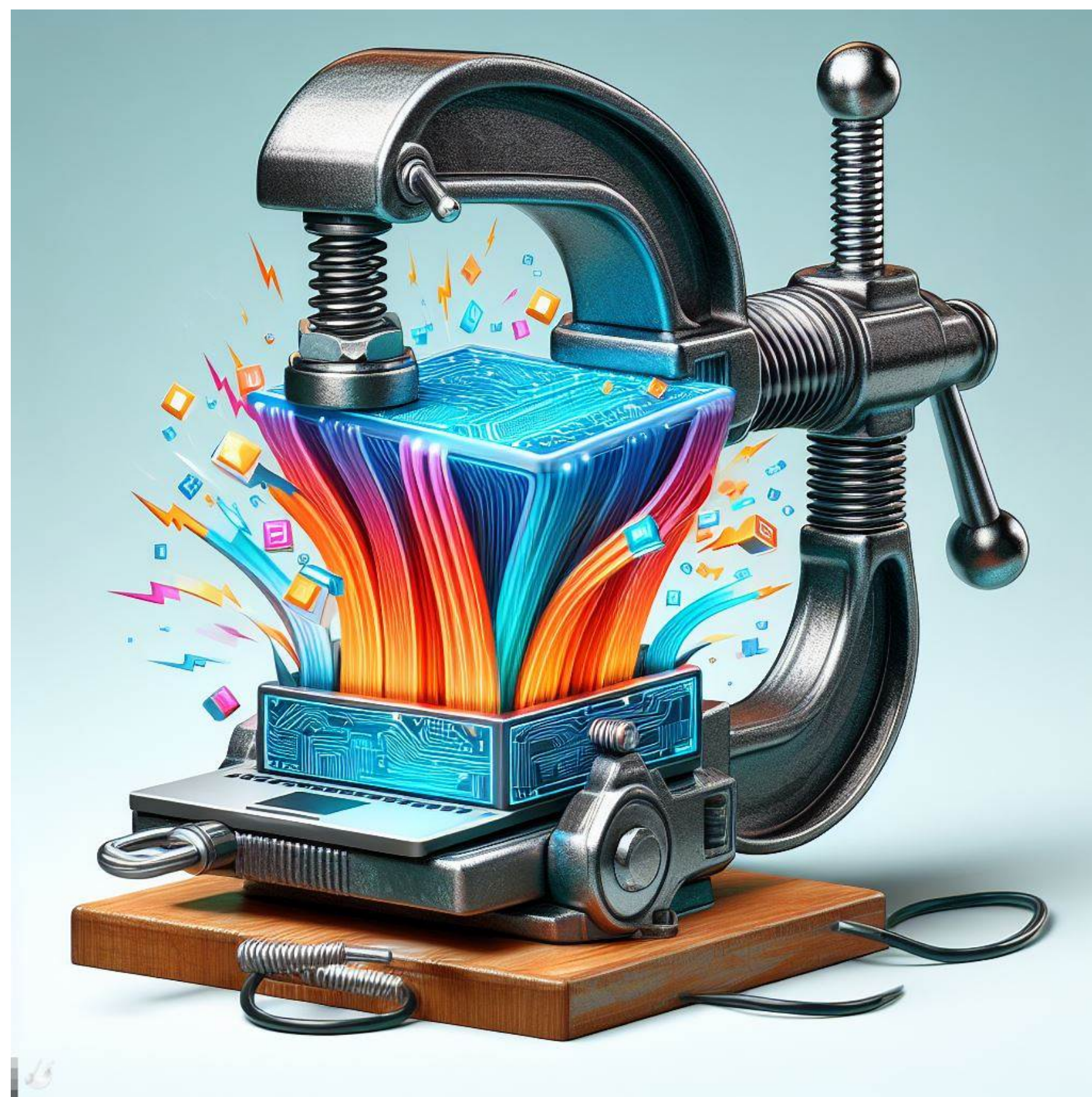
Multimedia data copression



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Introduction



Multimedia data compression is a foundational technology that underpins modern digital experiences. It enables the efficient storage of vast amounts of images, videos, and audio, reduces bandwidth requirements for faster transmission, and ensures seamless sharing across diverse devices and platforms. By optimizing file sizes without significantly compromising quality, it makes streaming, online communication, and multimedia-rich applications practical and accessible in our everyday digital lives.



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Why Compression Matters



- Compression is essential for two key reasons:
- First, it optimizes storage by significantly reducing file sizes, allowing large volumes of multimedia content to be stored more efficiently.
 - Second, it improves data transfer across networks, enabling faster downloads, smoother streaming, and more reliable communication, even over limited bandwidth.

Together, these benefits make compression a critical component of modern digital systems.



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Types of Compression

Multimedia compression can be classified into two main types: **lossy** and **lossless**.

Lossy compression reduces file sizes by selectively removing some data, which may slightly affect quality, but achieves much higher compression rates. In contrast, **lossless** compression preserves the original quality of the media entirely, ensuring that no data is lost, while still providing moderate file size reduction.

Both methods are essential, with lossy compression commonly used for streaming and web content, and lossless compression preferred for archival and professional applications.



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Image Compression

Many images hosted on the web are in a file format that uses lossy compression. This ensures the images load quickly and do not use too much bandwidth. Common examples of lossy compression methods include:

- **JPEG:** This file format, named after the Joint Photographic Experts Group, is perhaps the most widely used image compression method. More recent versions of JPEG include JPEG 2000 and JPEG XR, but many browsers do not support these formats.
- **WebP:** While WebP also supports lossless compression, it is more commonly used for lossy image compression. Google originally developed WebP to replace the JPEG, PNG, and GIF file formats.
- **High Efficiency Image Format (HEIF):** The High Efficiency Image Format (HEIF) is a type of container file for compressed images. Images compressed with this method are sometimes called HEIC files.



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Image Compression

Feature	JPEG	WebP	HEIF
Year introduced	1992	2010	2015
Compression type	Lossy (some lossless options)	Lossy & Lossless	Lossy & Lossless
File size	Moderate	Smaller than JPEG	Smaller than JPEG, often smaller than WebP
Image quality	Good, may degrade with high compression	Better quality at similar file size	Excellent quality at smaller file size
Transparency support	No	Yes	Yes
Animation support	No	Yes	Yes (HEIC sequence)
Adoption / support	Very high, universally supported	Growing, widely supported in browsers	Moderate, supported on Apple devices and some
Use cases	Web, photography, general images	Web, mobile apps, efficient media delivery	Mobile photography, iOS/macOS apps, modern storage



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Image Compression



 **2A1A2406.CR3** 24,2 MB

Modified: Today, 14:07

Add Tags...

▼ General:

Kind: RAW photo file
Size: 24 166 482 bytes (24,2 MB on disk)



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Image Compression



2A1A2406.jpg

184 KB

Modified: Today, 14:07

Add Tags...

▼ General:

Kind: JPEG image

Size: 183 983 bytes (184 KB on disk)



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PageSpeed Insights

PageSpeed Insights

[Copy Link](#) [Docs](#)

Report from Dec 9, 2025, 3:11:23 PM

<https://wroclaw.go.art.pl/>

Analyze

Mobile

Desktop



Discover what your real users are experiencing



No Data



Diagnose performance issues

65

Performance

85

Accessibility

96

Best Practices

92

SEO

65

Performance

Values are estimated and may vary. The [performance score is calculated](#) directly from these metrics. [See calculator.](#)



0-49



50-89



90-100

METRICS



First Contentful Paint

3.1 s



Largest Contentful Paint

5.5 s



Expand view



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Impact of Compression on Performance

- **Faster page loading** – smaller images and media reduce download times, improving user experience.
- **Optimized mobile networks** – lower bandwidth usage helps maintain performance on mobile and limited connections.
- **Reduced server load & costs** – smaller files mean less storage and lower data transfer expenses.



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Video Compression



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Video Compression Techniques

Video compression is shaped largely by industry standards such as **H.264/AVC** and **H.265/HEVC**, each offering distinct advantages. **H.264/AVC** remains the most widely used format for online streaming due to its strong balance between quality, efficiency, and broad device compatibility. In contrast, **H.265/HEVC** delivers significantly better compression ratios, making it ideal for ultra-high-definition content, including 4K and 8K video, where reduced bandwidth consumption is crucial. Newer codecs like **VP9** and **AV1** further push compression efficiency, targeting modern streaming platforms and low-latency applications. Collectively, these standards ensure that high-quality video can be delivered smoothly across a wide range of networks and devices.



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What is H.264?

- H.264, known as Advanced Video Coding (AVC), is a video codec that was released in 2003. Many video streaming services, such as YouTube and Netflix, use this codec today. H.264 was designed to provide high-quality video compression while maintaining a low bit rate, which is why it is ideal for streaming videos over the internet.



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What is H.265?

- H.265, also known as High Efficiency Video Coding (HEVC), is a video codec that was released in 2013. It is the successor to H.264 and was designed to provide better video compression while maintaining the same level of video quality. H.265 can achieve up to 50% more compression than H.264. So it is an ideal choice for high-resolution video formats.



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What is the difference between them?

	H.264/AVC	H.265/HEVC
Resolution	Bandwidth required	Bandwidth required
480p	1.5 Mbps	0.75 Mbps
720p	3 Mbps	1.5 Mbps
1080p	6 Mbps	3 Mbps
4K	32 Mbps	15 Mbps



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Video Compression

Feature	H.264 (AVC)	H.265 (HEVC)	H.266 (VVC)	AV1
Year introduced	2003	2013	2020	2018
Compression efficiency	Baseline	~50% better than H.264	~40–50% better than H.265	~20–30% better than H.265
Quality per bitrate	Good	Very good	Excellent	Excellent (especially in streaming)
Encoding complexity	Low	High	Very high	Very high
Decoding complexity	Low	Medium–high	Very high	High (improving with hardware)
Hardware support	Universal	Broad but not universal	Limited (emerging 2024–2025)	Growing fast (new GPUs, TVs, phones)
Licensing	Patent-encumbered	Heavy licensing	Heavy licensing	Royalty-free
Best use cases	Web, broadcast, older hardware	4K video, Apple ecosystem, Blu-ray UHD	4K/8K, VR/AR, professional workflows	Streaming (YouTube, Netflix), low-bandwidth delivery
Adoption speed	Mature & universal	Moderate	Slow	Fast-growing



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Video Compression

Resolution / Quality	H.264 (AVC)	H.265 (HEVC)	H.266 (VVC)	AV1
1080p (Full HD)	5–8 Mbps	3–5 Mbps	2–3 Mbps	2–3 Mbps
1440p (2K)	10–15 Mbps	6–9 Mbps	4–6 Mbps	4–6 Mbps
2160p (4K UHD)	25–35 Mbps	12–18 Mbps	8–12 Mbps	8–12 Mbps
4320p (8K UHD)	80–120 Mbps	40–60 Mbps	25–40 Mbps	25–40 Mbps



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Video Compression



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Audio Compression

Audio compression is a cornerstone of digital media, with formats like **MP3**, **AAC**, **Opus**, and **FLAC** widely used across various platforms. **MP3** offers a reliable balance between file size and sound quality, making it ideal for general listening. **AAC** provides superior efficiency and fidelity at similar bitrates, making it the preferred choice for streaming and mobile devices. **Opus** excels at low-bitrate, real-time applications such as VoIP and online conferencing, while **FLAC** delivers lossless compression for audiophiles and archival purposes, preserving every detail of the original recording.



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Audio Compression

Format	Typical bitrate	Quality
MP3	128–320 kbps	Good
AAC	128–256 kbps	Better than MP3 at same bitrate
Opus	64–128 kbps	Excellent for low-bandwidth streaming
FLAC	500–1000 kbps	Lossless, high quality



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Multimedia in mobile systems

Summary



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Why Multimedia Matters on Mobile

- Mobile devices are now **primary platforms for consuming media**.
- Multimedia (images, video, audio, interactive content) enhances:
 - **User engagement** (social media, messaging apps)
 - **Entertainment** (video streaming, mobile games, music)
 - **Education & communication** (e-learning, podcasts, video calls)

Challenges: limited storage, battery, network bandwidth, and processing power.

Takeaway: Efficient multimedia is key to **performance and user satisfaction**.



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Multimedia Formats – What We Use and Why

Images: JPEG, WebP, HEIF

- JPEG: widely supported, moderate compression
- WebP: better compression, supports transparency & animation
- HEIF: high efficiency, used on iOS, smaller file sizes

Audio: MP3, AAC, Opus, FLAC

- MP3: small size, good quality
- AAC: better efficiency, used in streaming & mobile
- Opus: excellent for low-bitrate streaming and calls
- FLAC: lossless, for audiophiles

Video: H.264, H.265, H.266, AV1

- H.264: universal compatibility
- H.265: better compression for HD/4K
- H.266: emerging, ultra-efficient for 8K/VR
- AV1: royalty-free, ideal for streaming

Why it matters: Choosing the right format balances **quality, file size, battery usage, and bandwidth.**



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Compression & Efficiency – How It Helps

Compression **reduces file size** → faster downloads and smoother playback.

Video: H.264 → H.265 → H.266 for higher resolutions with less bandwidth.

Audio: MP3/AAC/Opus reduce network load without significant quality loss.

Benefits:

- Lower **data consumption** → saves mobile data
- Faster **streaming and loading times**
- Less **battery and CPU usage** → longer device runtime

Example: Streaming a 4K video without compression could require 25–120 Mbps; compressed formats reduce this to 8–18 Mbps.



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Streaming on Mobile – Real-World Use

Video streaming apps: YouTube, Netflix, TikTok

- Use H.264/H.265/AV1 to stream HD and 4K efficiently

Audio streaming apps: Spotify, Apple Music

- Use AAC, Opus, MP3 to deliver high-quality sound with low bandwidth

Why it matters: Users expect **instant access, minimal buffering, and high quality**

Challenge: Mobile networks are often unstable → compression ensures **consistent experience**



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Multimedia in Interactive Apps

Mobile games & AR/VR apps rely heavily on multimedia assets:

- 3D models, textures, animations, sound effects

Optimization is crucial for:

- **Battery life**
- **CPU/GPU load**
- **Memory usage**

Why it matters: Without efficient multimedia, apps can **lag, crash, or drain battery quickly.**



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Summary and Questions



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Useful links

Compression: Crash Course Computer Science #21: <https://www.youtube.com/watch?v=OtDxDvCpPL4>

Data compression: <https://www.khanacademy.org/computing/computers-and-internet/xcae6f4a7ff015e7d:digital-information/xcae6f4a7ff015e7d:data-compression/a/file-compression-introduction>



“What we know is a drop, what we
don't know is an ocean.”

Isaac Newton
